SpaceX Falcon 9
First-stage landing prediction

Applied Data Science Capstone
Course dataset analysis

Executive summary

60 of 90 first stages were labeled successful (66.7%).

CCAFS SLC 40 hosted 55 launches, the most in this sample.

A decision tree had the highest mean training cross-validation accuracy (75%). Its holdout accuracy was 77.8% (14/18).

The sample is small and historical. Results describe associations, not causes.

Source: IBM Skills Network course dataset

Introduction

Falcon 9 is designed to recover and reuse its first stage.

Question: Can mission information help predict whether the first stage lands successfully?

Analysis covers launch sites, payloads, orbit, time and reuse, followed by classification.

The target describes landing outcome, not whether the payload mission succeeded.

Source: IBM Skills Network course dataset

Dataset and scope

90 Falcon 9 records from 4 June 2010 through 5 November 2020.

Class 1: successful first-stage landing or controlled ocean outcome.
Class 0: unsuccessful or no landing, including no attempt.

Course-prepared payload data includes imputed values.

A separate 101-row course CSV supports the SQL section. Counts across the two files are not directly comparable.

Source: IBM Skills Network course dataset

Data collection

Course-provided CSV reflects the project's SpaceX API collection and data-wrangling workflow.

A separate course CSV supplies the launch table for SQL exploration.

For this reproduction, both datasets were downloaded from IBM Skills Network and saved locally.

Data, code, and model outputs can be rerun from the accompanying project folder.

Source: IBM Skills Network course dataset

Data wrangling

Kept launch site, date, orbit, payload mass and flight-related features.

Checked class counts and launch-site distribution.

For modeling: split before training, then impute, scale numeric variables and one-hot encode categories inside each training fold.

The supplied dataset has already filled some missing payload values.

Source: IBM Skills Network course dataset

Exploratory analysis methods

Grouped launches by site, orbit and year.

Calculated landing successes divided by launches in each group.

Examined launch counts alongside rates to avoid treating a small group as conclusive.

Compared reused and new boosters as an association, not a controlled experiment.

Source: IBM Skills Network course dataset

SQL analysis methods

Loaded the separate SpaceX CSV into a local SQLite table.

Used SELECT, WHERE, GROUP BY and aggregation to inspect sites, payloads and outcomes.

Preserved source site labels, including CCAFS LC-40 and CCAFS SLC-40.

The SQL CSV has 101 records, versus 90 in the modeling file.

Source: IBM Skills Network course dataset

Interactive visual analytics methods

Folium: placed markers at the coordinates of each launch site and attached site-level counts.

Plotly Dash: added launch-site and payload-range filters with an outcome chart and payload scatterplot.

The local Dash app applies both filters before updating its figures.

Source: IBM Skills Network course dataset

Predictive analysis methods

Target: first-stage landing class.

Features: payload, flights, reuse count, block, reused status, orbit and launch site.

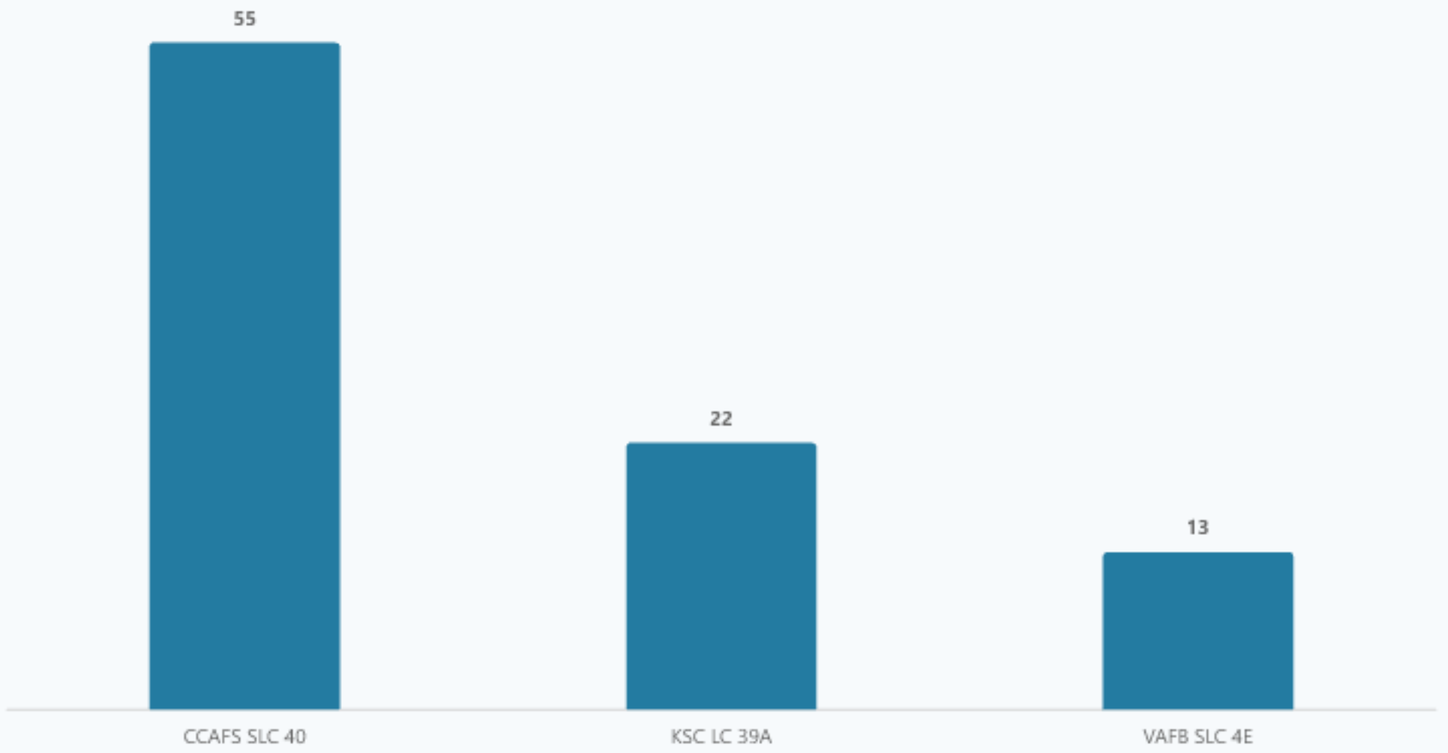
Stratified 80/20 split: 72 training records and 18 test records.

Compared a majority baseline with logistic regression, SVM, decision tree and KNN using four-fold training cross-validation.

No hyperparameter search was performed.

Source: IBM Skills Network course dataset

CCAFS SLC 40 hosted 55 of 90 launches

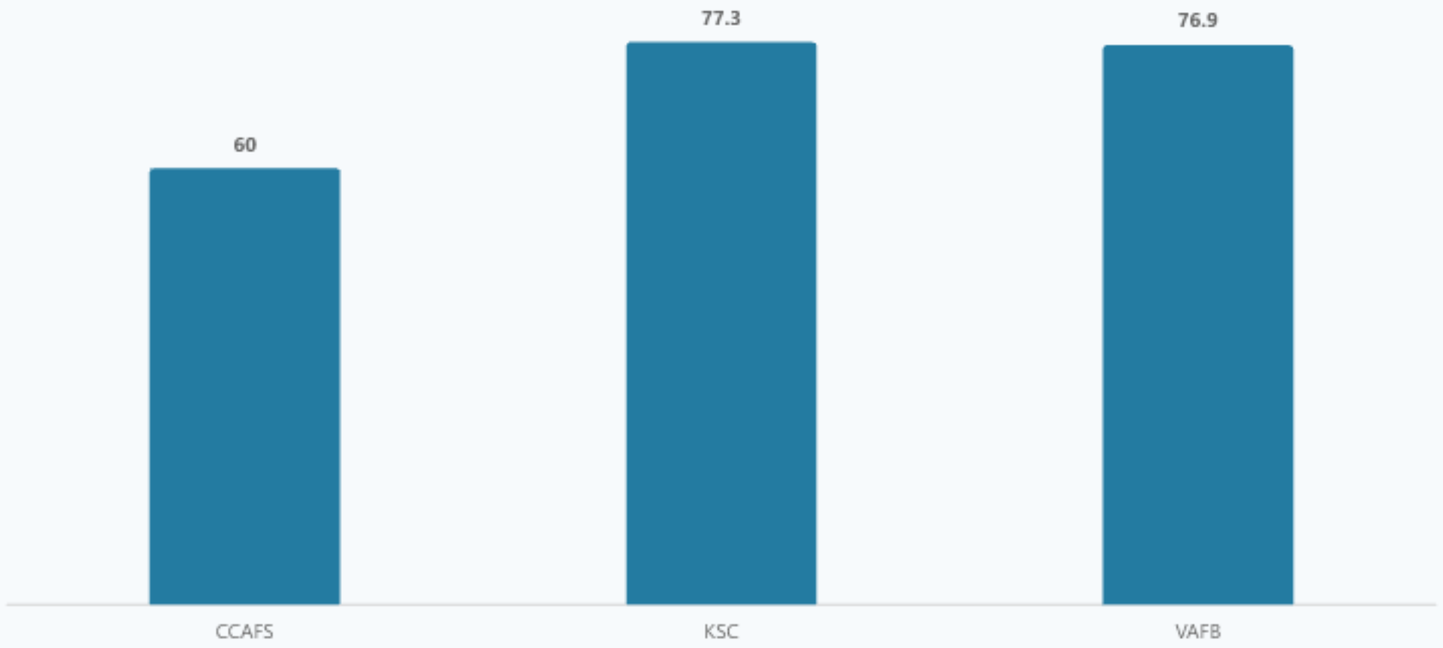


Launch counts differ substantially by site.

The CCAFS sample includes 33 successful landings out of 55 launches.

Source: IBM Skills Network course dataset (90 records)

Landing success differed across launch sites

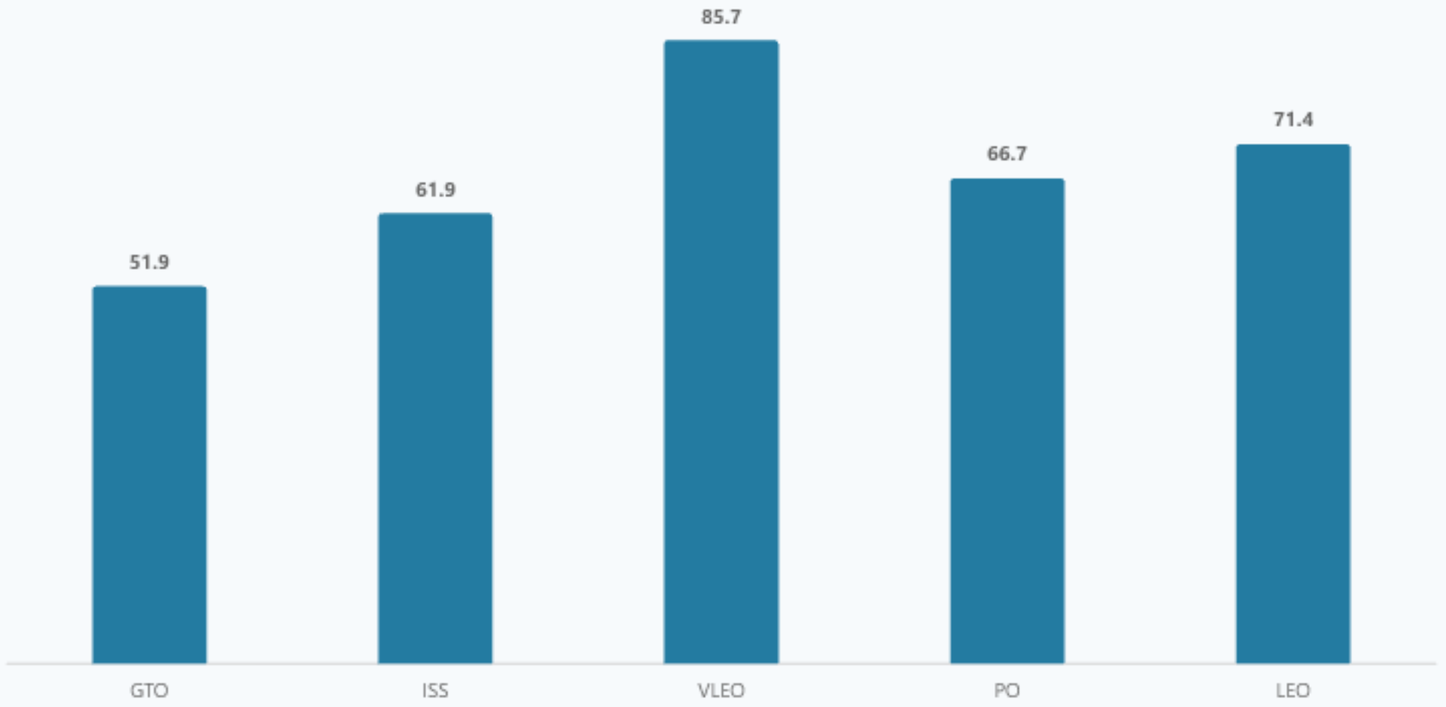


CCAFS: 33 / 55
KSC: 17 / 22
VAFB: 10 / 13

These historical differences do not establish that site caused the outcome.

Source: IBM Skills Network course dataset (90 records)

Success rates varied by orbit

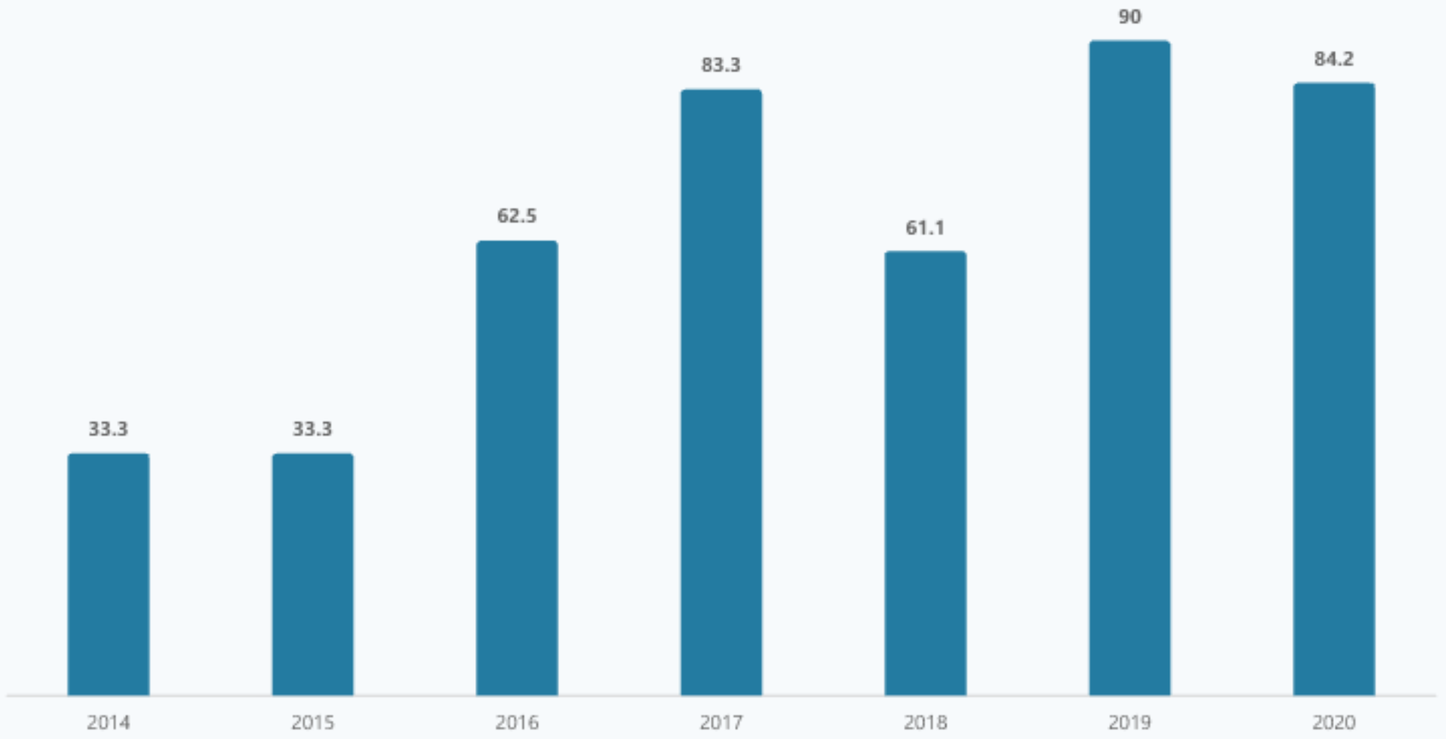


GTO: 14 / 27
ISS: 13 / 21
VLEO: 12 / 14

Orbit groups differ in size and launch conditions.

Source: IBM Skills Network course dataset (90 records)

Later years generally had higher success rates

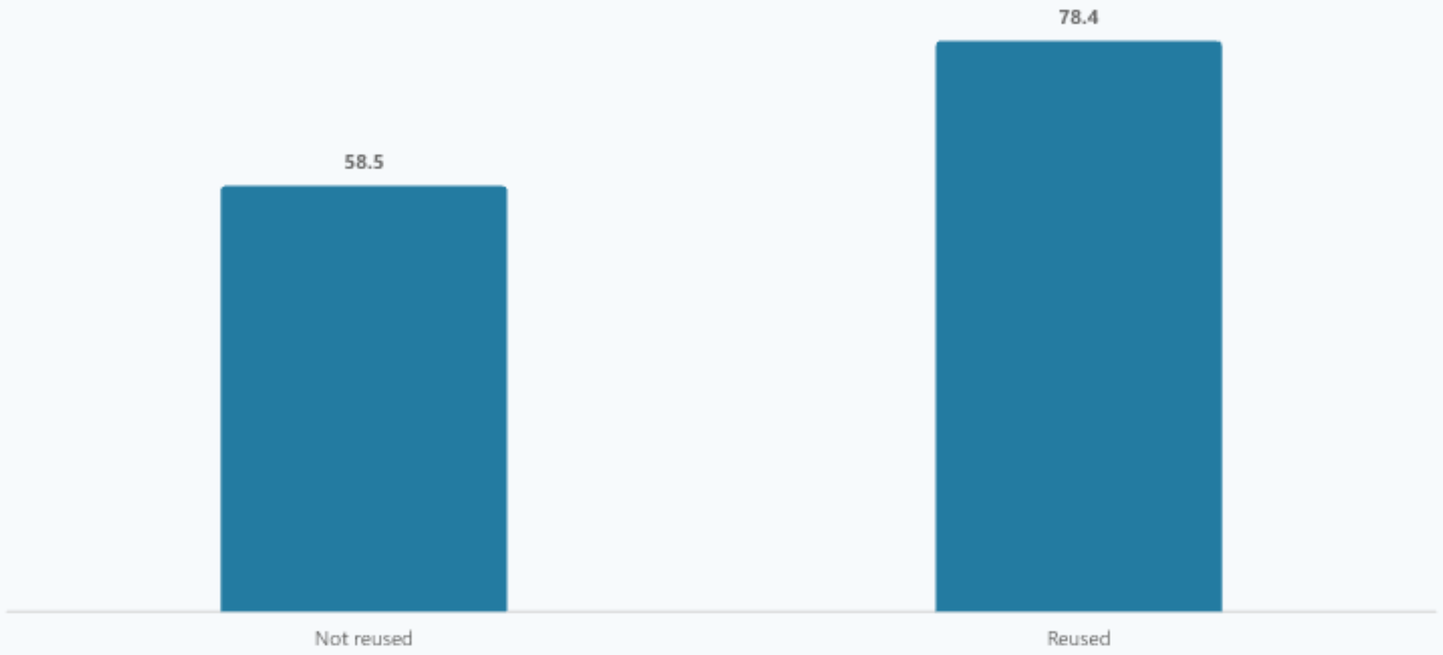


2014: 2 / 6
2017: 15 / 18
2020: 16 / 19

Rates reflect a selected dataset and small yearly samples.

Source: IBM Skills Network course dataset (90 records)

Reused boosters had a higher observed landing rate



Not reused: 31 / 53
Reused: 29 / 37

Booster design and launch year may also differ across groups. The comparison is not causal.

Source: IBM Skills Network course dataset (90 records)

SQL results: launch sites

```
SELECT Launch_Site, COUNT(*) FROM launches
GROUP BY Launch_Site ORDER BY COUNT(*) DESC;
```

CCAFS SLC-40: 34 CCAFS LC-40: 26
KSC LC-39A: 25 VAFB SLC-4E: 16

Labels in the SQL source are retained as recorded.

Source: IBM SQL course CSV (101 records)

SQL results: NASA payload and Falcon 9 v1.1

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM launches
WHERE Customer = 'NASA (CRS)';
```

Total NASA (CRS) payload: 45,596 kg

F9 v1.1 average payload: 2,534.67 kg
Calculated from rows with booster version beginning “F9 v1.1”.

Source: IBM SQL course CSV (101 records)

SQL results: landing and payload

First recorded successful ground-pad landing: 22 Dec 2015.

Maximum payload mass in this SQL file: 15,600 kg. The maximum occurs on multiple rows.

Landing outcome categories include 14 explicitly labeled “Success (drone ship)”.

The separate 90-row modeling data uses a different outcome coding.

Source: IBM SQL course CSV (101 records)

SQL results: mission outcomes and boosters

After trimming whitespace, the 101 mission outcomes are:
99 success, 1 in-flight failure, 1 success with unclear payload status.

Four drone-ship successes carried between 4,000 and 6,000 kg:
B1022 (4,696), B1026 (4,600), B1021.2 (5,300), B1031.2 (5,200).

Mission outcome and first-stage landing outcome are separate fields.

Source: IBM SQL course CSV (101 records)

SQL results: early landing outcomes

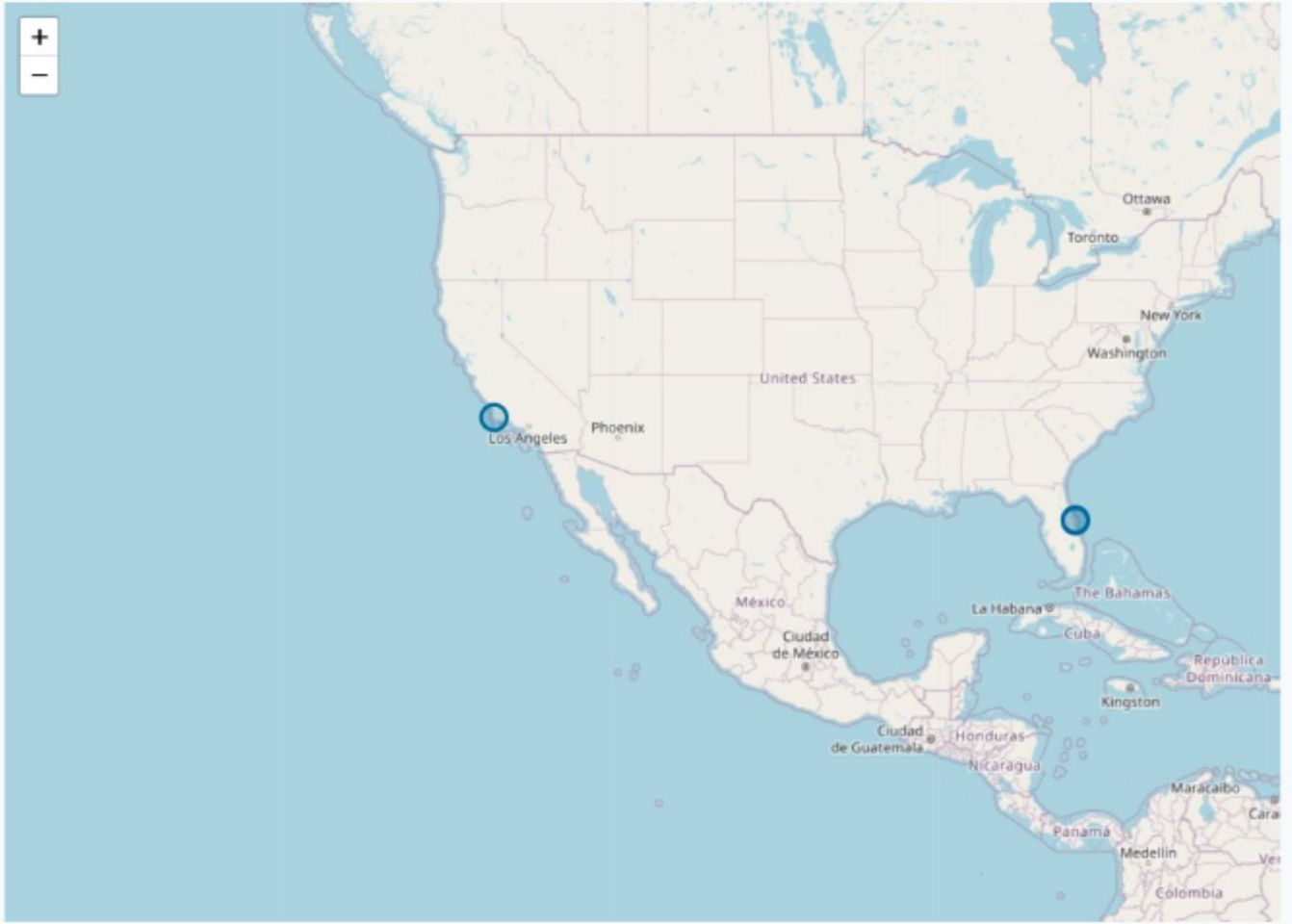
Two drone-ship landing failures in 2015:
10 Jan: B1012, CCAFS LC-40
14 Apr: B1015, CCAFS LC-40

From 4 Jun 2010 through 20 Mar 2017:
10 no attempt, 5 drone-ship successes and 5 drone-ship failures.

First five CCA-prefixed rows in the SQL file are CCAFS LC-40 launches.

Source: IBM SQL course CSV (101 records)

Folium map: launch sites



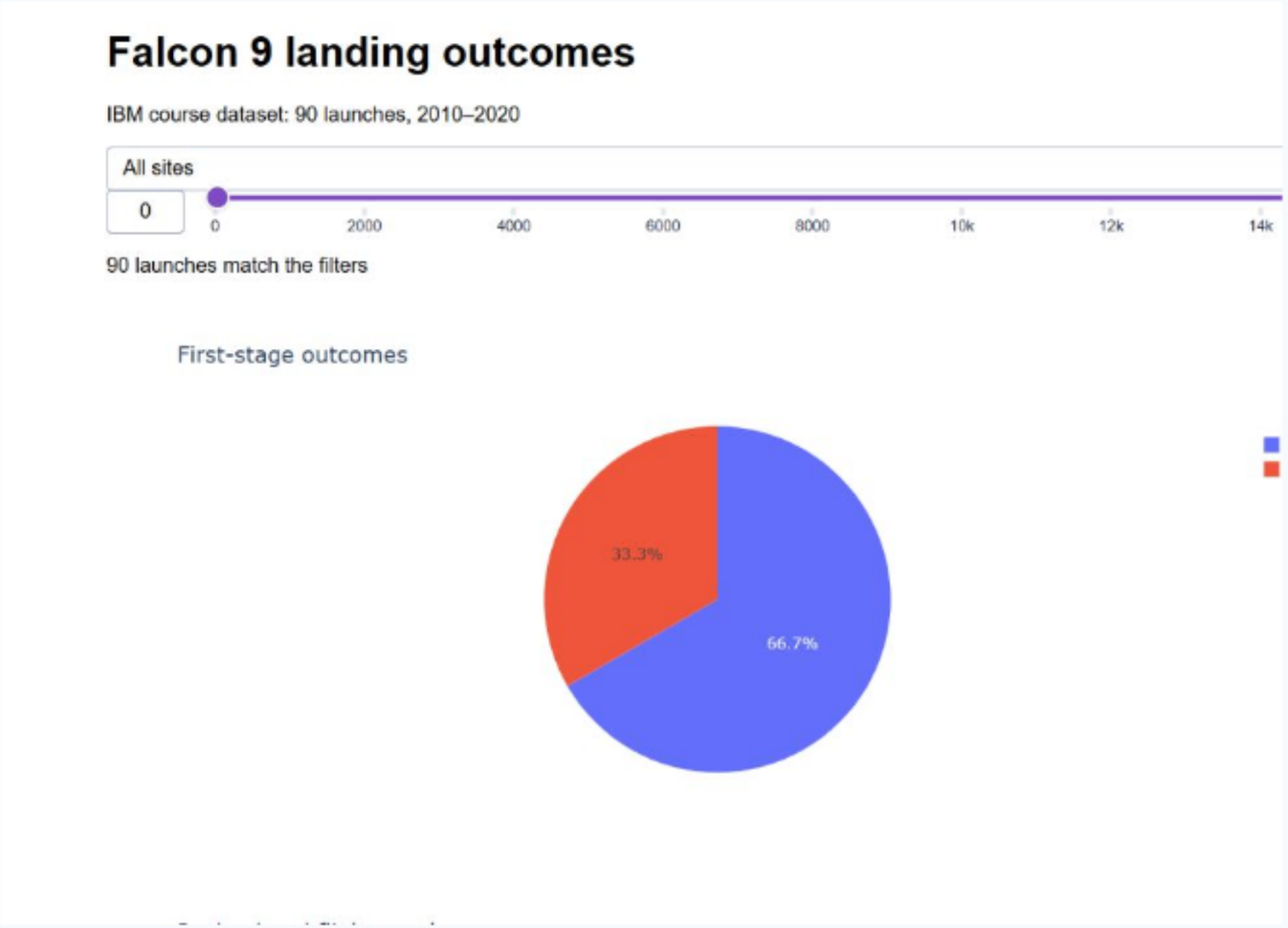
The dataset places two sites in Florida and one in California.

Markers show location; the two Florida sites overlap at this zoom.

Interactive map popups show launch counts and successful landings.

Data: IBM course dataset. Map © OpenStreetMap contributors.

Dash dashboard: exploring outcomes



The default view includes all 90 records.

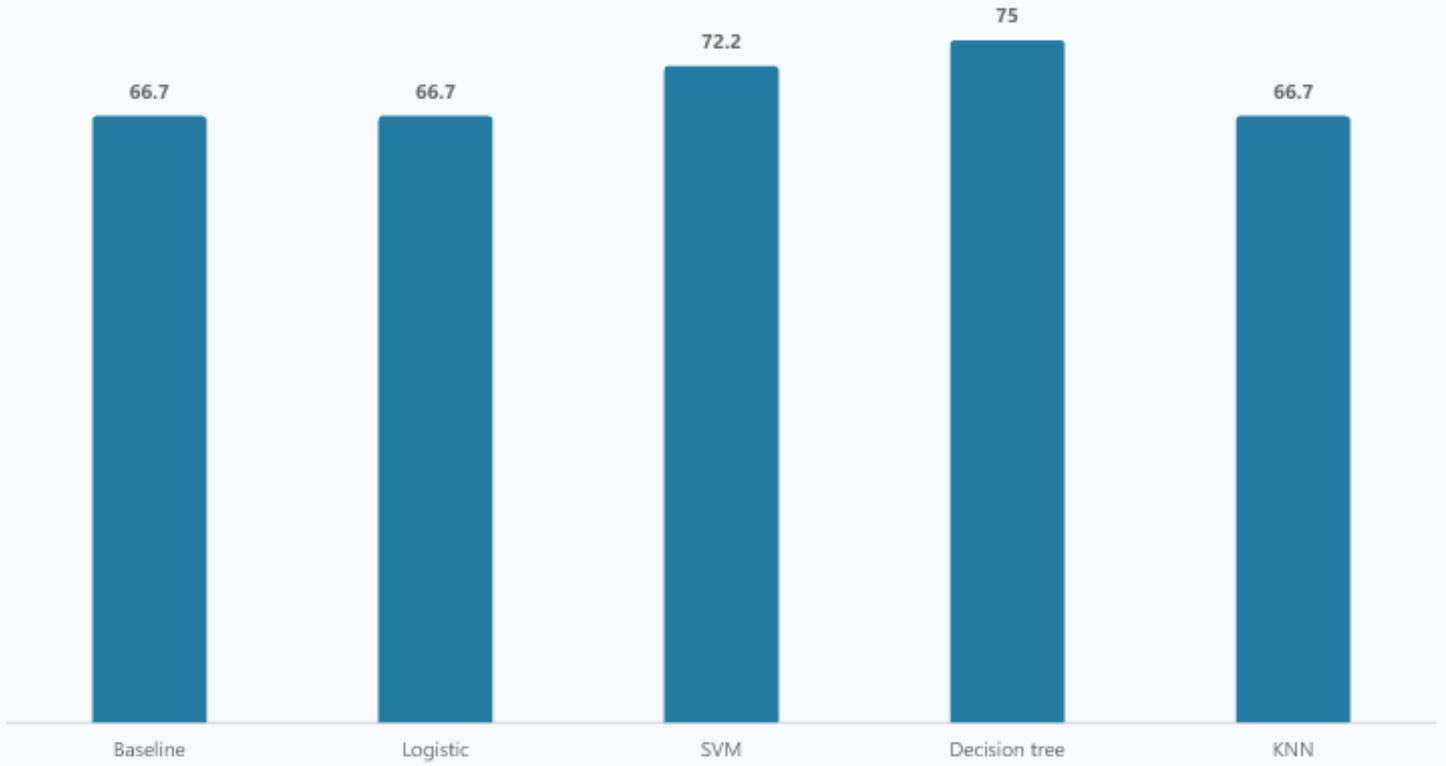
60 successful landings: 66.7%.

Site and payload filters update the outcome chart and scatterplot together.

Static screenshot of a locally run app.

Source: IBM Skills Network course dataset

Decision tree led training cross-validation



Four-fold cross-validation on 72 training records.

The decision tree had the highest mean score.

The differences are uncertain with this small sample.

Source: IBM Skills Network course dataset (90 records)

Decision tree: holdout results

Selected using training cross-validation, the decision tree correctly classified 14 of 18 holdout records (77.8%).

Confusion matrix	Predicted no	Predicted yes
Actual no landing	3	3
Actual successful landing	1	11

The model missed half of the six non-successes in this test split.

Source: local analysis, 18-record stratified holdout; seed 42

Model comparison and uncertainty

Holdout accuracy: logistic regression 88.9%, SVM 77.8%, decision tree 77.8%, KNN 77.8%, majority baseline 66.7%.

The tree was selected on training cross-validation, not on the holdout result.

A single test split of 18 records is unstable. Further time-based testing on newer launches is needed before operational use.

Source: local analysis; fixed settings, no hyperparameter search

Conclusion

Landing success increased in later years within this course sample.

Site, orbit and reused status show differences, but timing and mission characteristics may confound them.

The selected decision tree surpassed the majority baseline on this holdout, while missing 3 of 6 non-successes.

Next: test on later launches and verify that each predictor is available before launch.

Source: IBM Skills Network course dataset

Sources and reproducibility

IBM Skills Network: dataset_part_2.csv (90 launches)
cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_2.csv

IBM Skills Network: Spacex.csv (101 SQL records)
cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/labs/module_2/data/Spacex.csv

Code: analyze.py, sql_extra.py and visual_analytics.py

Repository: https://github.com/sabakan0123/spacex-falcon9-capstone

Folium map tiles © OpenStreetMap contributors